

Vishak Shashikumar

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TECHNICAL SKILLS

AI & Machine Learning: LLM application development (Claude, OpenAI, Ollama/Mistral), retrieval-augmented generation (RAG), vector search (FAISS), embeddings, prompt engineering, prompt-injection hardening, LLM cost observability, classification, forecasting (SARIMAX), anomaly detection

Languages & Data: Python, SQL, TypeScript, Pandas, NumPy, scikit-learn, PostgreSQL, SQLAlchemy 2.x, Alembic, Pydantic, ETL/ELT pipelines, data modeling, statistical analysis

Analytics & BI: Tableau, Power BI, Streamlit, KPI dashboards, executive reporting, cohort and trend analysis, Matplotlib, Recharts

Backend & Platform: FastAPI, Flask, REST API design, Next.js/React, pytest, Docker, Docker Compose, Kubernetes, Git/GitHub, AWS (S3, SageMaker, DynamoDB)

PROFESSIONAL EXPERIENCE

Ipser Labs Remote
AI Engineer Intern Jan 2026 – Jul 2026

- Built and secured FastAPI services powering the LLM features of a consumer application, adding vector-database retrieval (RAG) to ground model responses in real application data
- Hardened LLM-facing endpoints against prompt injection with input sanitization and structured prompt isolation, validated by an automated pytest suite of adversarial payloads
- Delivered end-to-end LLM cost observability: per-call token and cost instrumentation, persistence to an indexed usage table, an authenticated endpoint aggregating spend by user, day, and operation, and configurable daily-budget alerting
- Eliminated silent data loss in that pipeline by replacing fire-and-forget async writes with awaited transactional commits, so every model call was recorded before the request returned

Balaji AutoTech Pvt Ltd Mysuru, India
Data Analyst Jan 2022 – Jun 2024

- Engineered scalable ETL pipelines to extract, clean, and unify manufacturing data, delivering 95%+ accuracy across operational analytics workflows
- Developed anomaly detection models and interactive Tableau and Power BI dashboards that cut equipment downtime 15% through real-time fault monitoring
- Deployed containerized analytics services with Docker and Kubernetes, giving teams reliable API access to aggregated production metrics
- Partnered with cross-functional engineering teams to optimize data flow, improving pipeline performance and operational visibility for process owners

PROJECTS

Second Brain for Life — Personal AI Knowledge & Journaling Platform Jul 2026 – Present

- Built a fully local RAG system over a personal knowledge base: FAISS search across sentence-transformer embeddings feeds a self-hosted Mistral model (Ollama), streamed token-by-token from FastAPI at zero inference cost, with no user data leaving the machine
- Modeled a normalized PostgreSQL schema (typed SQLAlchemy 2.x, Pydantic-validated REST) serving aggregation queries to a TypeScript/Next.js dashboard (React, Recharts) for journaling and retrieval trends; packaged with Docker Compose

CareCast — Pediatric ICU Forecasting System Oct 2025 – Dec 2025

- Built a SARIMAX forecasting pipeline on large-scale HHS hospital data predicting 7–60-day pediatric ICU utilization, with automated cleaning, anomaly handling, and capacity-shock simulations
- Shipped an interactive Streamlit platform with KPI cards, sparkline trends, and state-level ICU maps; open-sourced under the MIT License

Personal Assistant Chatbot — RAG on AWS Mar 2025 – Jul 2025

- Engineered a RAG assistant that answers questions over a user's own documents, pairing an OpenAI model with a document-embedding pipeline and a Streamlit chat interface
- Ran it on AWS: S3 for document storage, DynamoDB for authenticated session persistence, SageMaker for embedding and sentiment-analysis workloads

EDUCATION

Northeastern University Boston, MA, USA
Master of Science in Applied Machine Intelligence Dec 2026

JSS Science & Technology University Mysuru, India
Bachelor of Engineering in Computer Science May 2021